

AWS RDS (MariaDB) SETUP

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Project: AWS RDS Setup

Objective

- Login AWS
- Project Purpose

Architecture Diagram

- Structure
- Explanation

Tools & Services Used

- AWS RDS
- MariaDB
- Amazon EC2
- Security Group
- Linux Terminal

Implementation Steps

- Command used
- Output

Conclusion

- Summary
- Learning Outcome

1. OBJECTIVE-

- Login to AWS Console.
- Search RDS.
- Open the RDS Dashboard

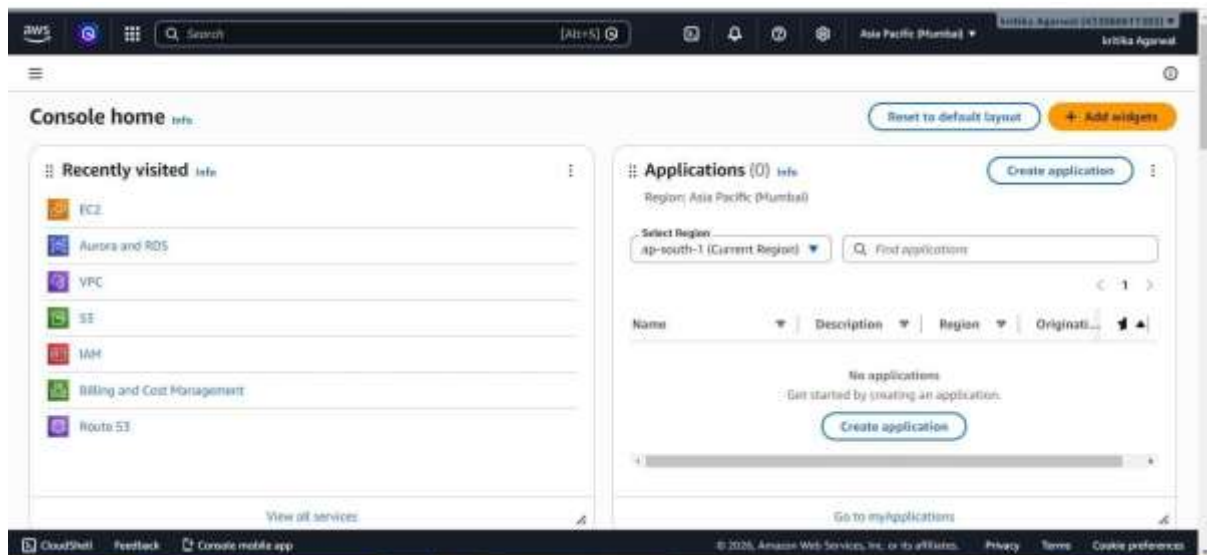


FIG: AWS CONSOLE HOME PAGE

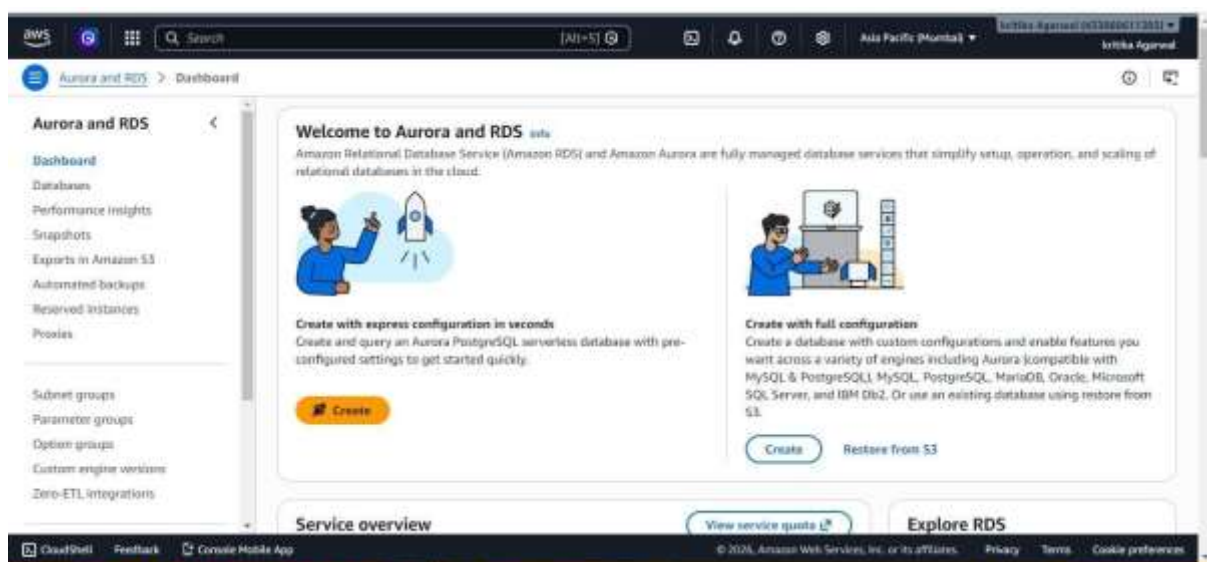


FIG:RDS DASHBOARD

2. ARCHITECTURE DIAGRAM-

1. User:

- User AWS Management Console me login karta hai.
- Yahan se RDS service ko manage karta hai.

2. AWS Management Console:

- AWS ka web interface hai.
- Iske through naya RDS instance create aur configure kiya jata hai.

3. Amazon RDS (MariaDB):

- AWS ki managed database service hai.

- Ye database setup, backup, monitoring aur maintenance automatically handle karti hai.

4. RDS Database Instance:

- Ye actual MariaDB server hota hai.
- Isme database create hota hai aur data store hota hai.

5. Database (Tables):

- Database ke andar tables hoti hain.
- Tables me application ka data rows aur columns ke form me save hota hai.

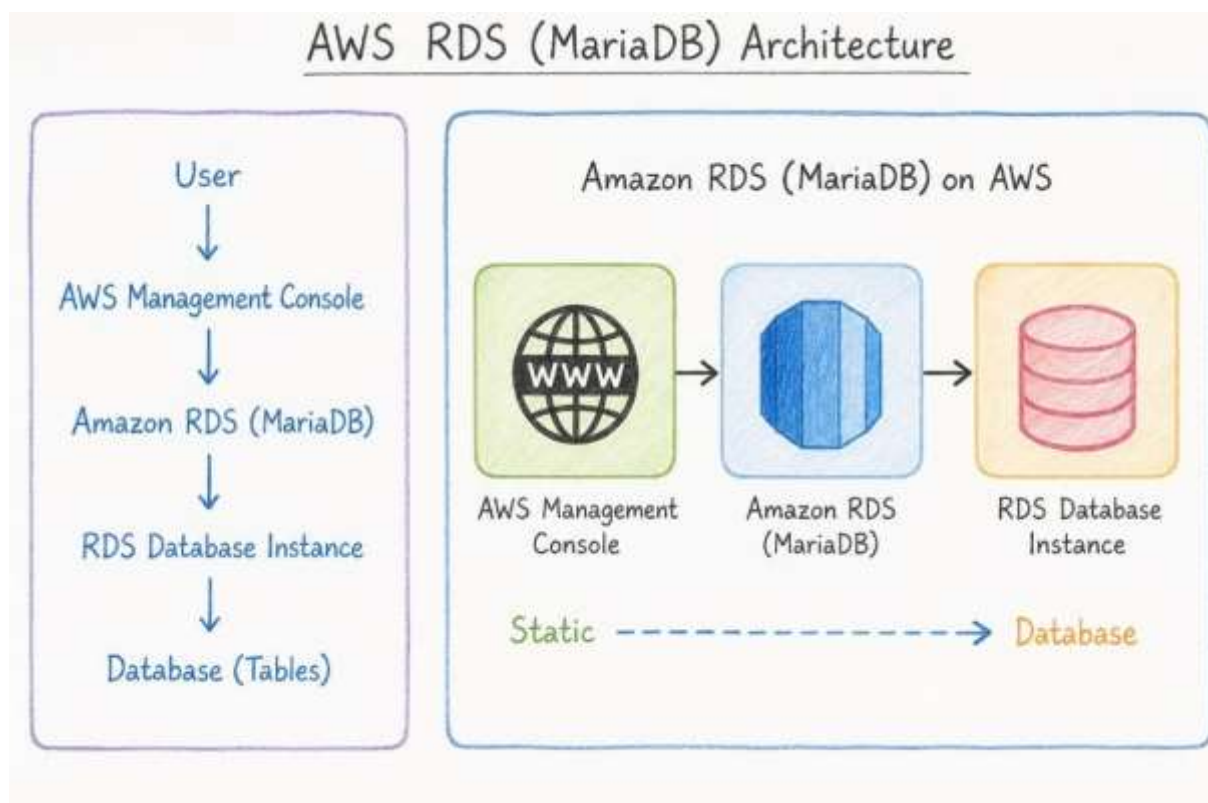


FIG: RDS ARCHITECTURE

3. TOOLS & SERVICES USED-

STEP1: Launch EC2 Instance-

- Open EC2 Dashboard.
- Launch EC2 Instance.
- Select Amazon Linux.
- Create key Pair.
- Security Group.
- EC2 instance running.

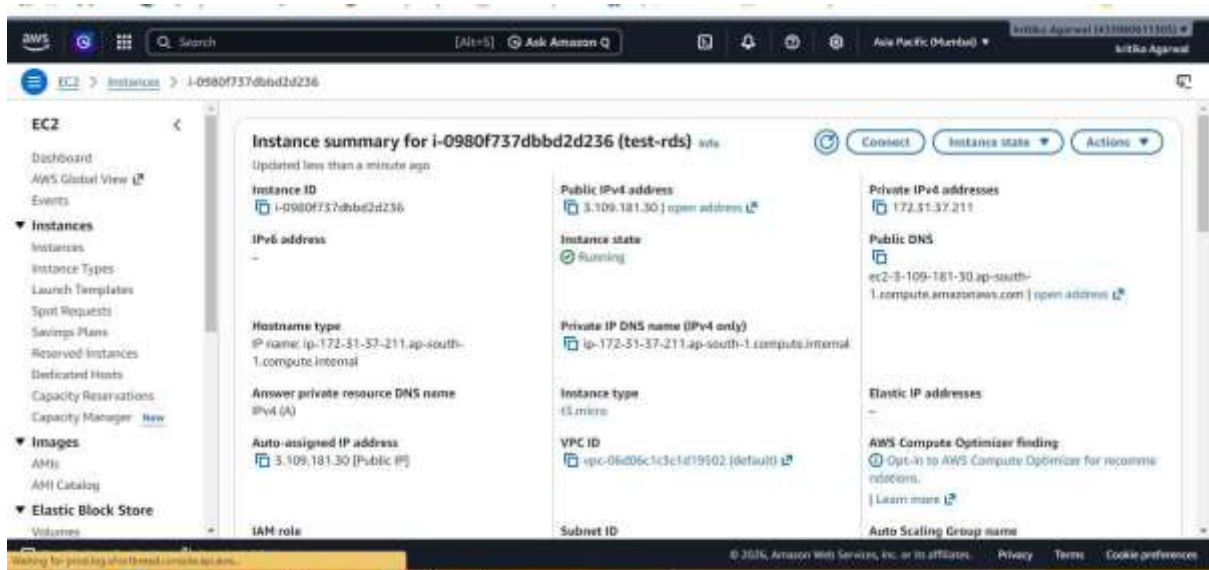


FIG:INSTANCE RUNNING

STEP2:Open RDS-

- Login to AWS Console
- Search RDS
- Click Create Database

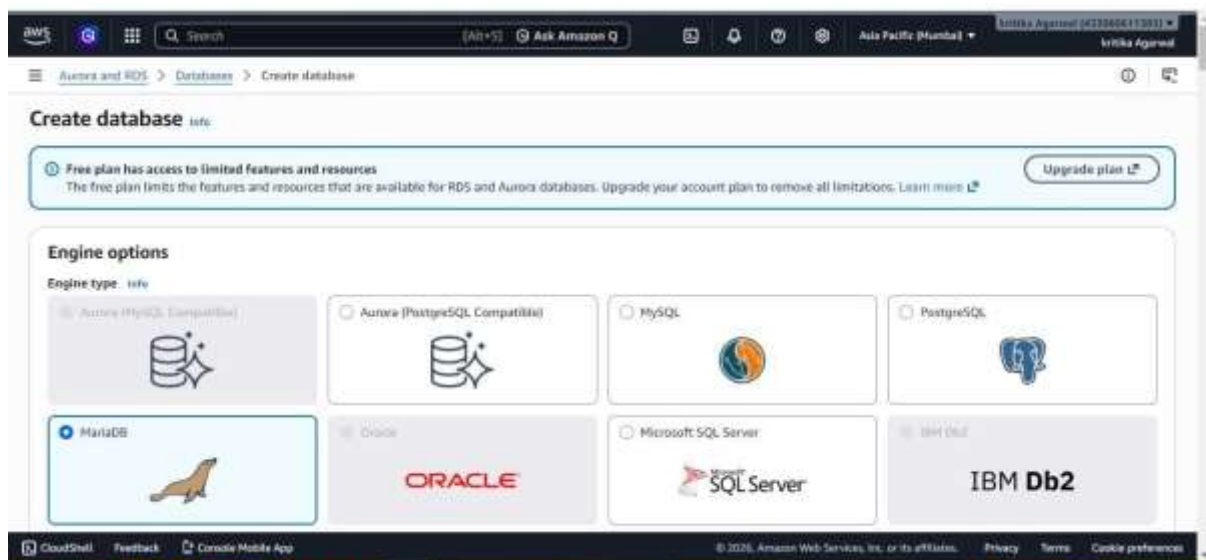


FIG: CREATE DATABASE

STEP3: Choose Database Creation-

- Select Standard Create.
- Engine Type → MariaDB.
- Choose the latest/default version.

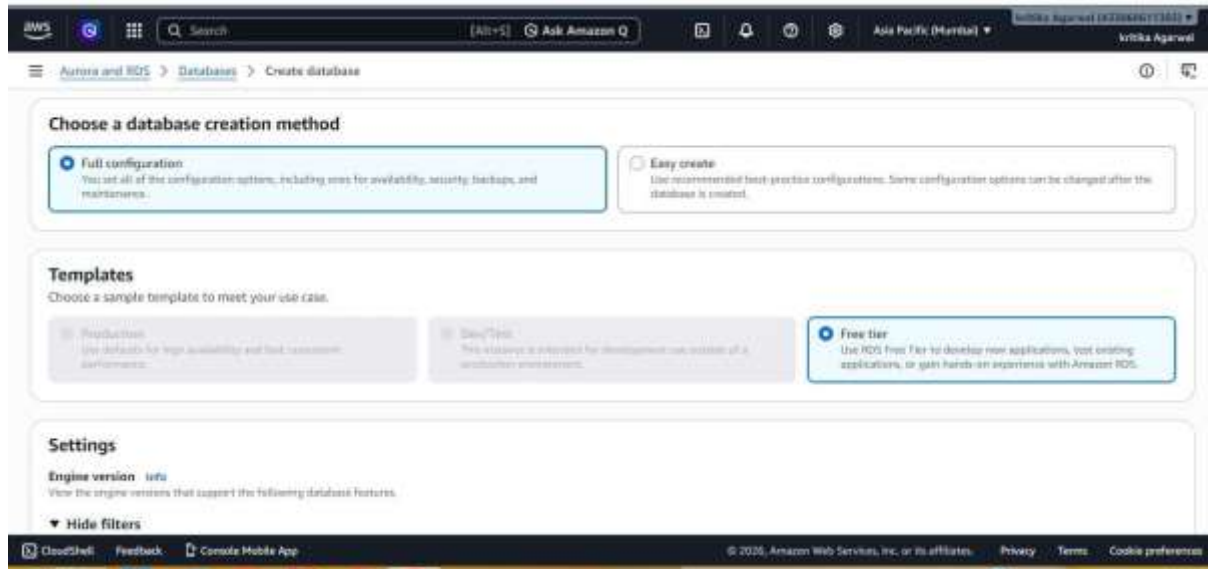


FIG: CHOOSE DATABASE

STEP4: Configure Database-

- DB Instance Identifier → database01
- Master Username → admin
- Create a strong password.

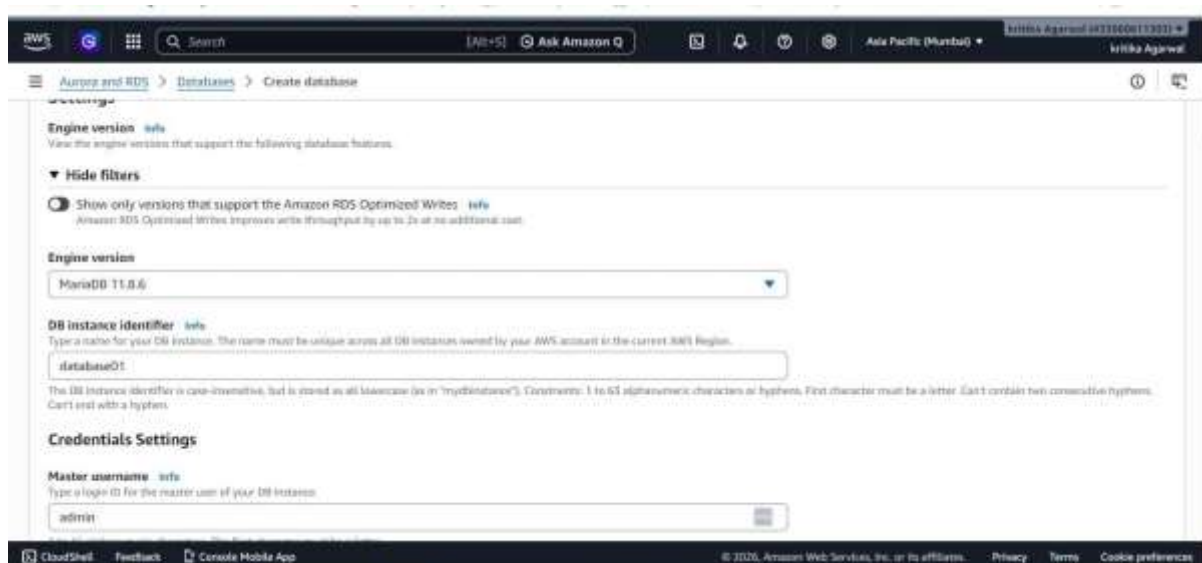


FIG-1:DB IDENTIFIER

The screenshot shows the 'Create database' page in the AWS Management Console. The 'Master username' field is set to 'admin'. Under 'Credentials management', 'Managed in AWS Secrets Manager' is selected. The 'Master password' field is masked, and a 'Password strength' bar is shown. The 'Confirm master password' field is also masked. The page includes a top navigation bar with the AWS logo, search bar, and user profile, and a bottom footer with links to CloudShell, Feedback, and the Console Mobile App.

FIG:MASTER USERNAME

STEP5:Instance Configuration-

- DB Instance Class → db.t3.micro
- Storage → Default (20 GB).

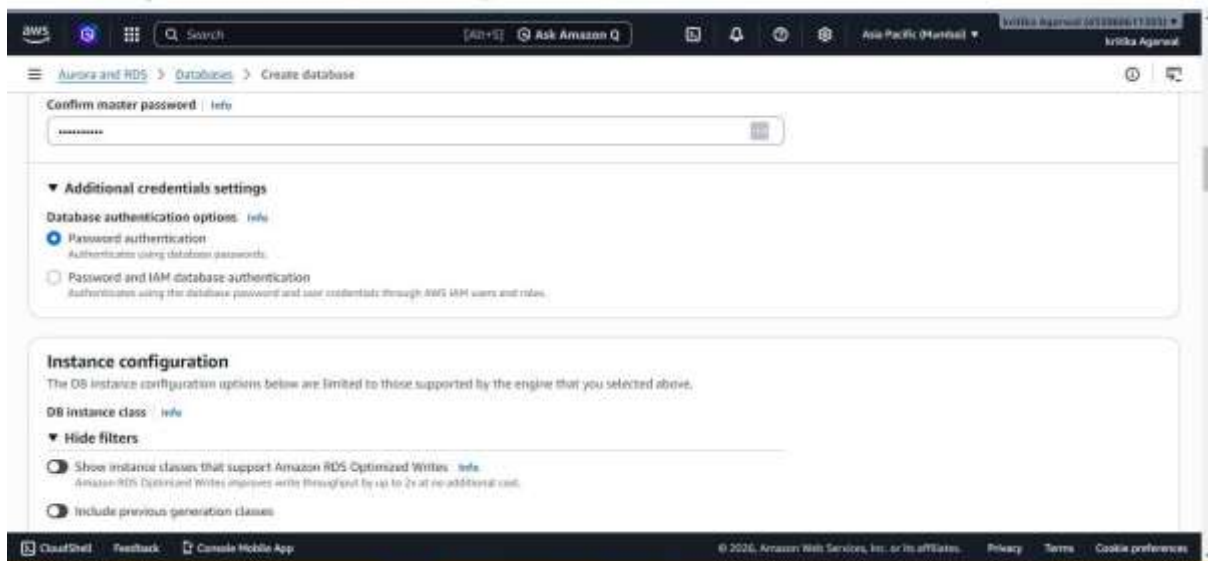


FIG-1:INSTANCE CONFIGURATION

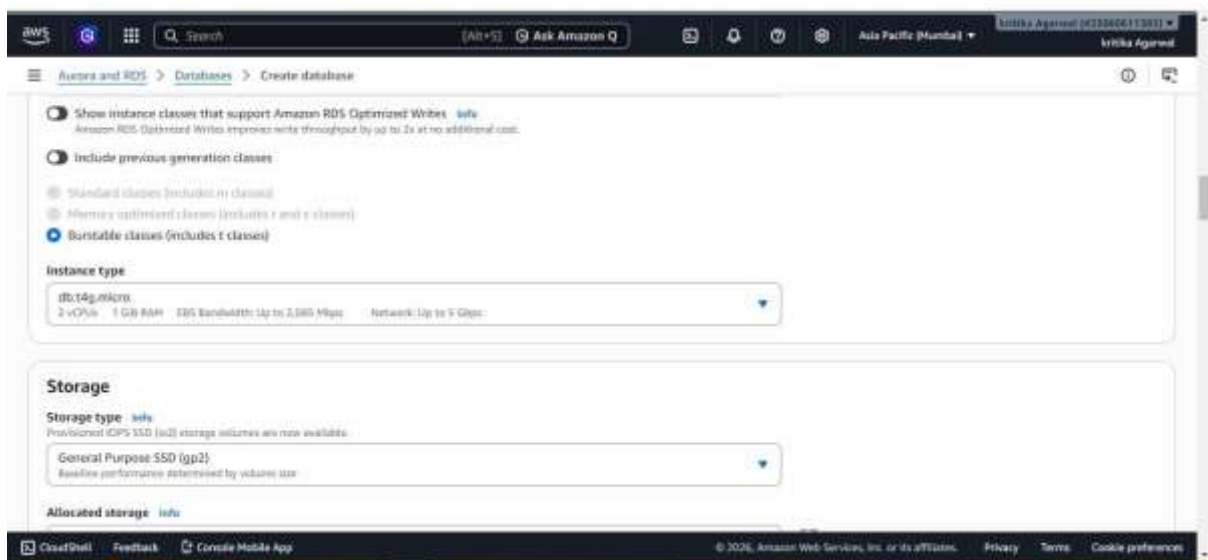


FIG-2: INSTANCE CLASS

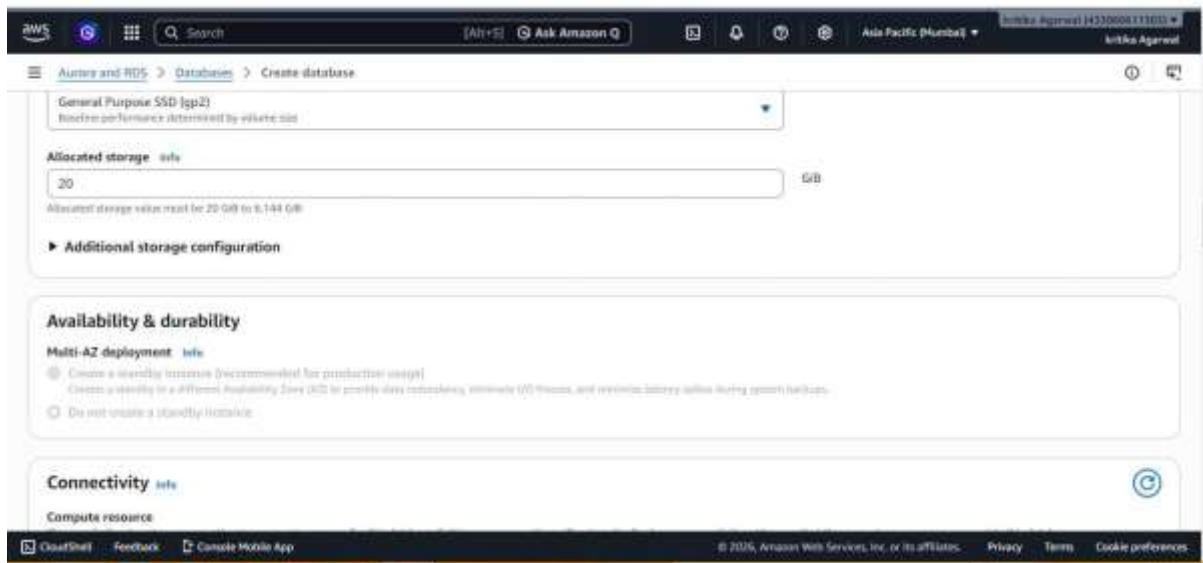


FIG-3:STORAGE

STEP6:Security Group-

- Enabled Public Access.
- Allowed Port 3306 in the Security Group.

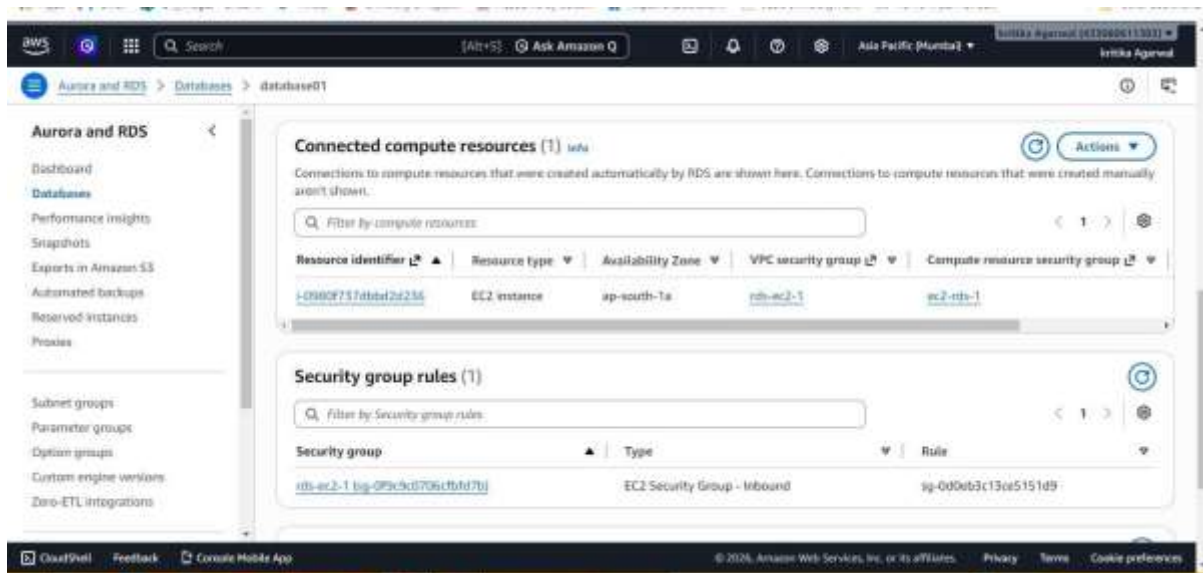


FIG:SECURITY GROUP

STEP7:Connectivity-

- Select VPC.
- Select/Create Security Group.
- Allow Port 3306 (MySQL/MariaDB).

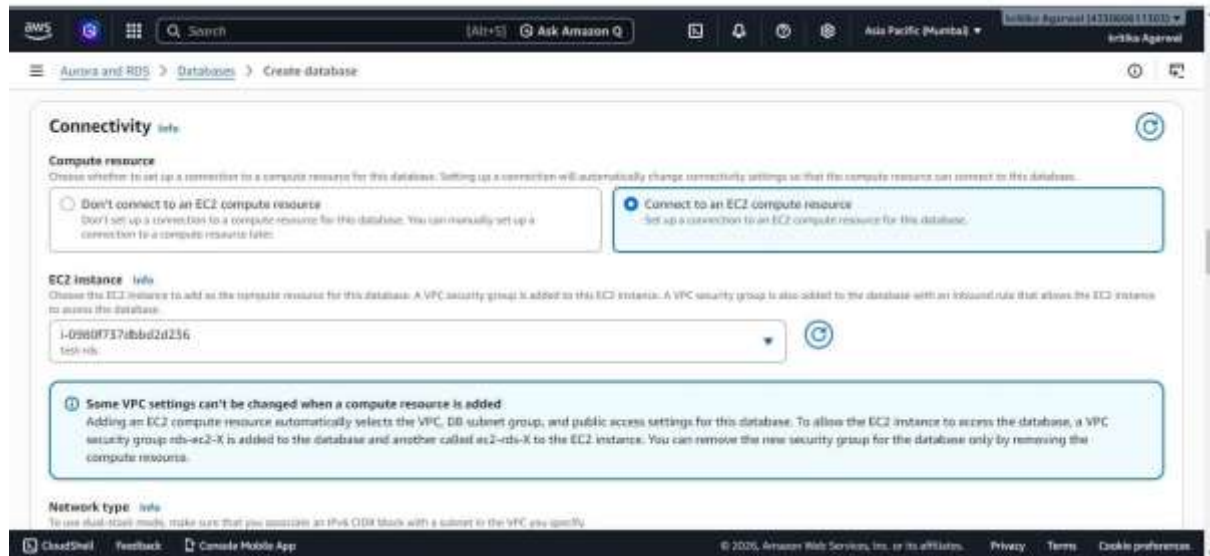


FIG-1:CONNECTIVITY

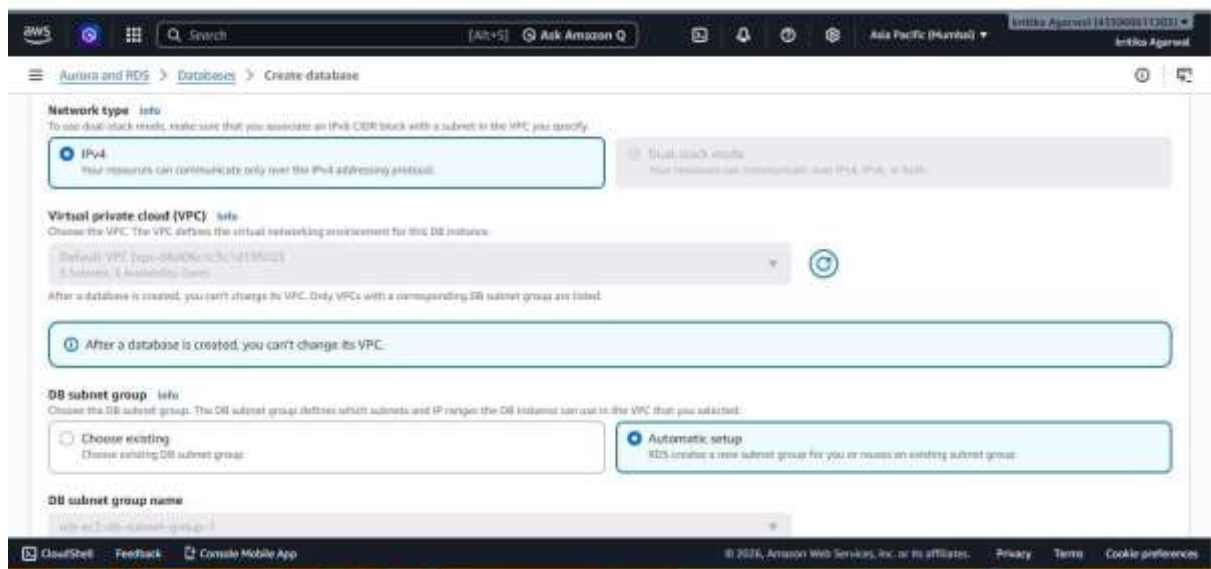


FIG-2:VIRTUAL PRIVATE CLOUD

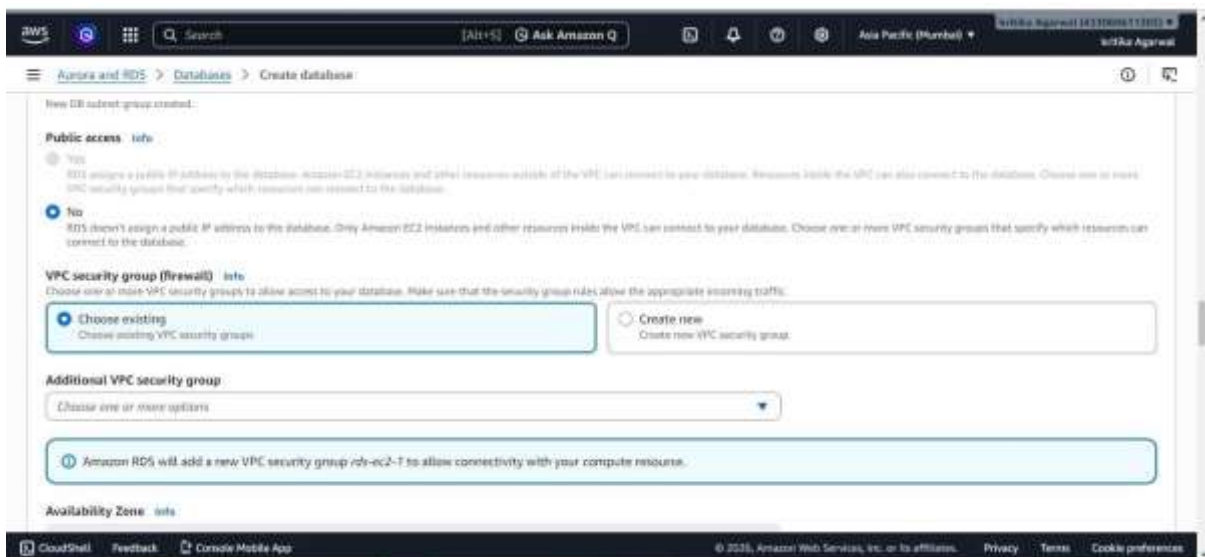


FIG-3:PUBLIC ACCESS

STEP8:Additional Configuration-

- Keep remaining settings as default.
- Click Create Database.

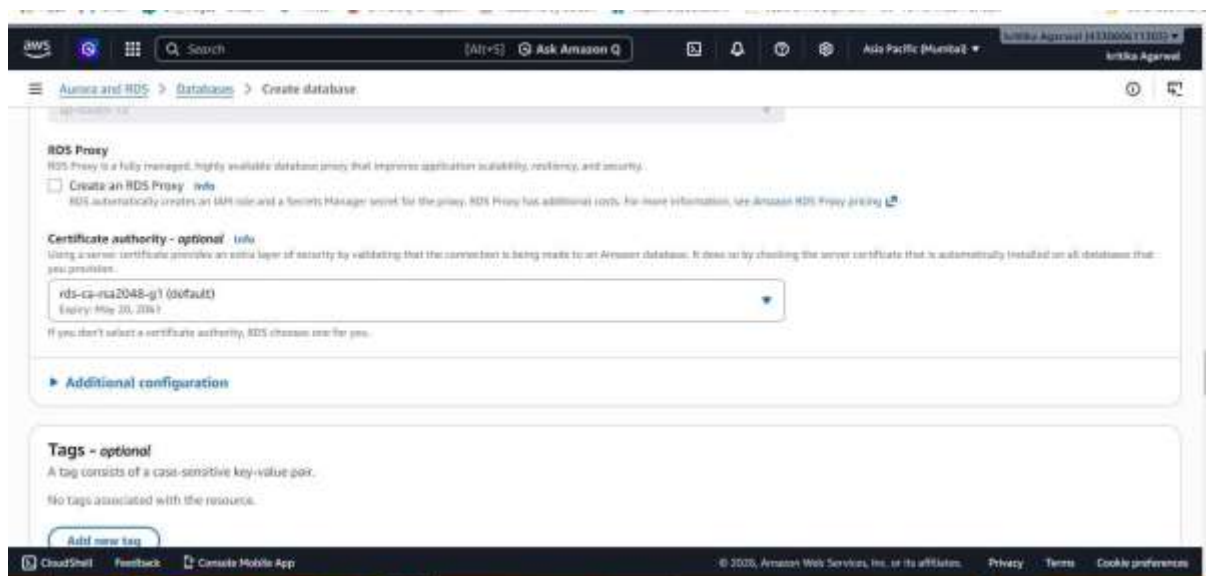


FIG-1:ADDITIONAL DATA

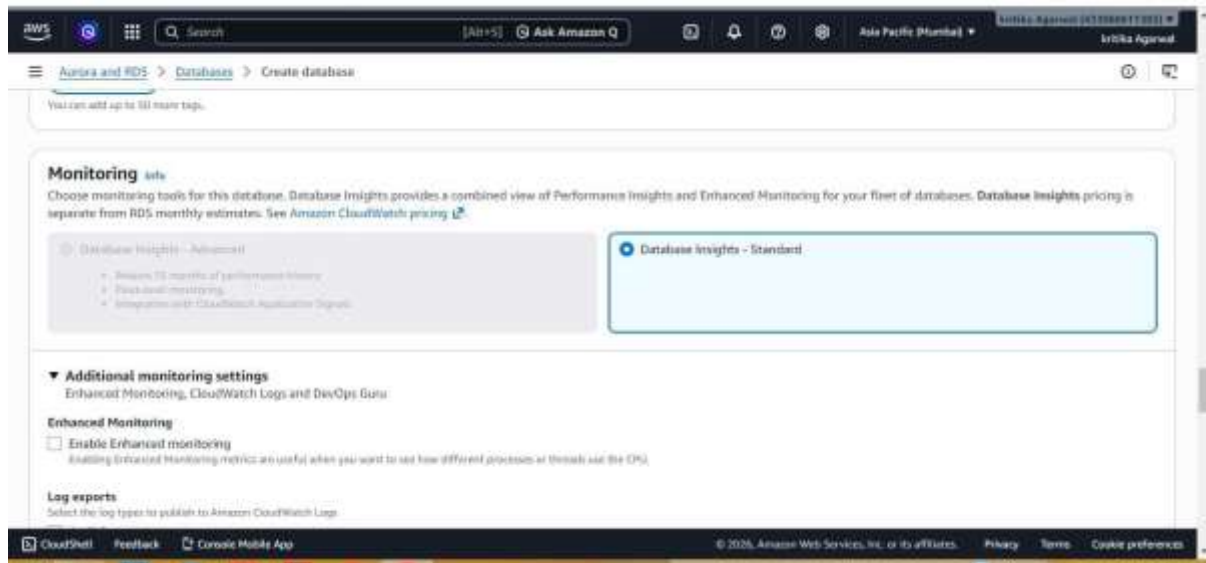


FIG-2:MONITORING

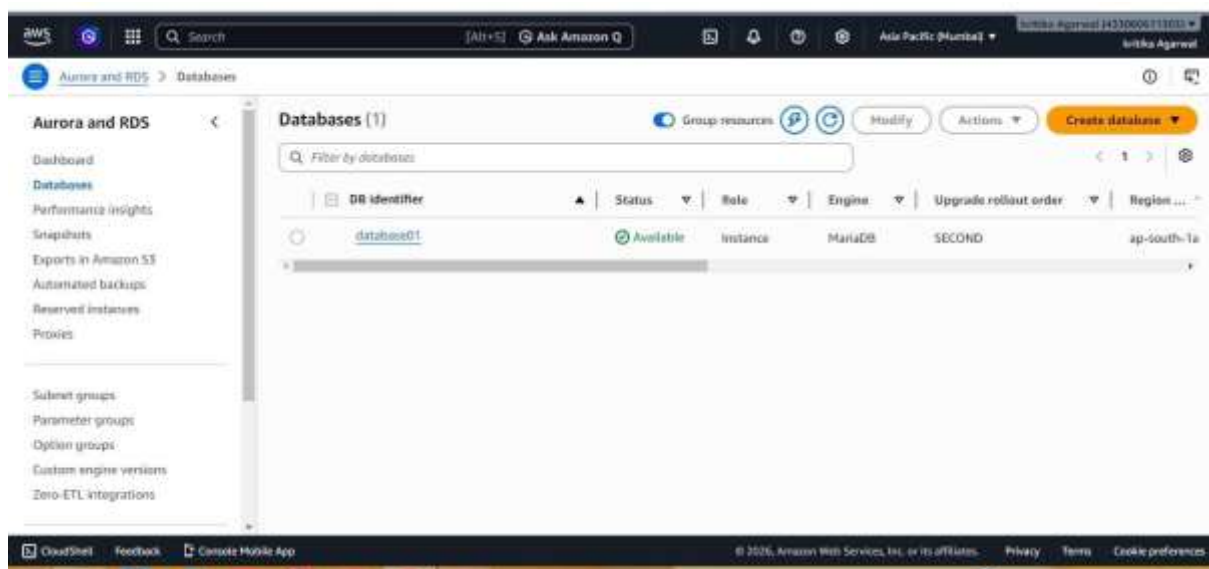


FIG-3:CREATE DATABASE

STEP9:Wait For Availability-

- Wait until the database status becomes Available.
- Copy the RDS Endpoint.

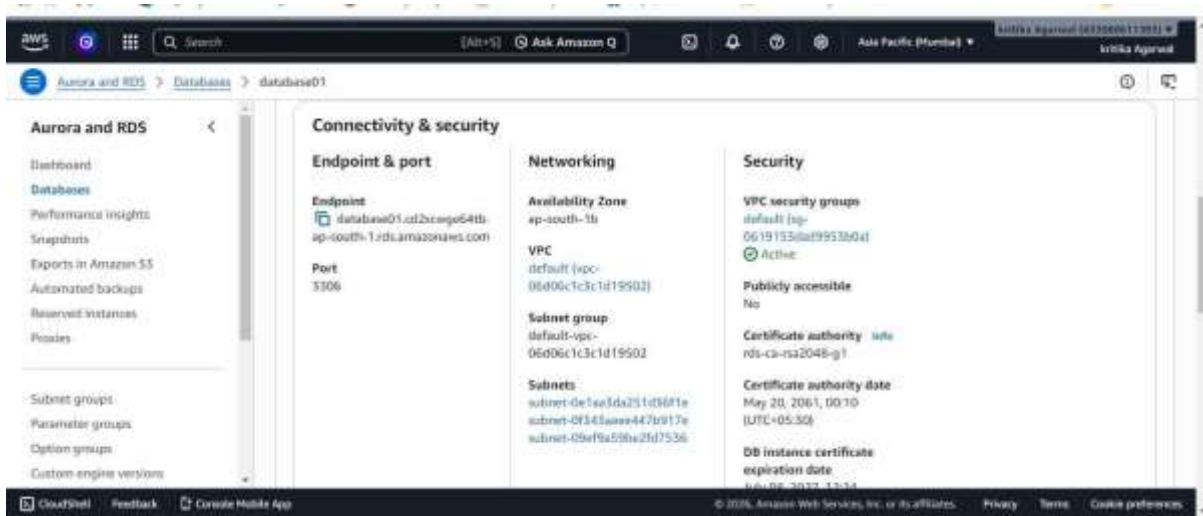


FIG: COPY ENDPOINT

4. IMPLEMENTATION STEPS:

STEP1: Connect from EC2 Terminal-

- Install MariaDB/MySQL client:-
- `dnf install -y mariadb-server`
- `Systemctl status mariadb`
- `Systemctl enable --now mariadb`

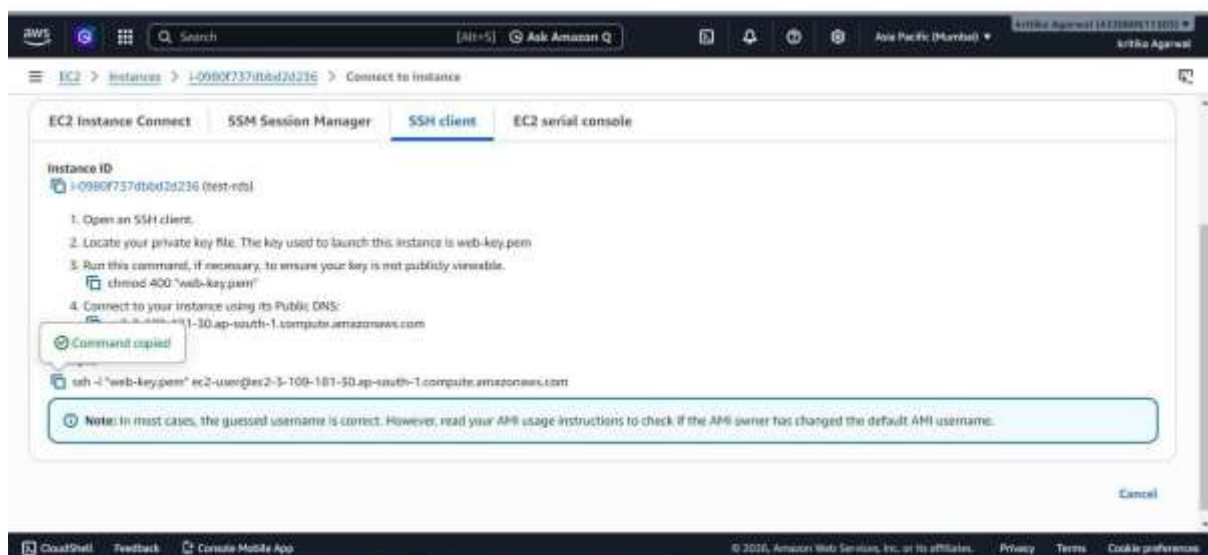


FIG-1:CONNECT SSH

```
root@RDS:~#
The RDS client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://rhc.rh/registration
Last login: Tue Jul 7 09:04:14 2020 from 157.46.99.246

[ec2-user@RDS ~]$
[ec2-user@RDS ~]$ sudo -i
[ec2-user@RDS ~]$ hostnamectl set-hostname RDS
[ec2-user@RDS ~]$ sudo
[ec2-user@RDS ~]$ sudo systemctl --no-reload enable mariadb-server
[ec2-user@RDS ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset; disabled)
   Active: active (running) since Tue 2020-07-07 09:04:14 UTC; 20min ago
   Process: 41404403374473006047a5b34831
   Main PID: 2220 (mariadbd)
   Status: "Taking your SQL requests now..."
   Tasks: 9 (limit: 4156)
   Memory: 113.8M (peak: 318.1M)
   CPU: 70ms
   CGroup: /system.slice/mariadb.service
           └─2220 /usr/bin/mariadbd --basedir=/usr

Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: you need to be the system 'mysql' user to connect.
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: After connecting you can set the password, if you would need to be
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: able to connect as any of these users with a password and without sudo
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: Please report any problems at https://mariadb.org/t/r/a
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: The latest information about MariaDB is available at https://mariadb.org/.
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: Consider joining MariaDB's strong and vibrant community!
Jul 07 09:05:04 RDS mariadb-prepare-db-dir[2270]: https://mariadb.org/get-involved
Jul 07 09:05:04 RDS [mariadbd][2220]: mariadb.service: Referenced but shared environment variable evaluation to an empty string: MARIADB_OPTS, _MARIADB_CLUSTER
Jul 07 09:05:04 RDS systemd[1]: Started mariadb.service - MariaDB 10.11 database server.
[ec2-user@RDS ~]$
```

FIG-2: ENABLE MARIADB SERVICE

STEP2:Connect to RDS-

- mysql -h <RDS-Endpoint> -u admin -p
- Enter the master password.

```
root@RDS:~#
[ec2-user@RDS ~]$ mysql -h database-01.c2scvge64tch.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
mysql: [Warning] Access denied for user 'admin'@'172.31.37.211' (using password: YES)
[ec2-user@RDS ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Root@12345';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
bye
[ec2-user@RDS ~]$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

FIG:RDS ENDPOINT (SUCCESSFUL LOGIN)

STEP3:Create Database & Verify-

- CREATE DATABASE database01;
- SHOW DATABASES;

```
root@RDS:~#
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '?' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
# rows in set (0.001 sec)

MariaDB [(none)]> exit
bye
[root@RDS ~]# mysql -u root
ERROR 1045 (42000): Access denied for user 'root'@'localhost' (using password: NO)
[root@RDS ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 55
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '?' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
# rows in set (0.000 sec)

MariaDB [(none)]>
```

FIG:DATABASE VERIFY

OUTPUT:

- MariaDB RDS instance created successfully.
- EC2 connected to the RDS database.
- Database database01 created successfully.

5.CONCLUSION:

SUMMARY-

The AWS RDS MariaDB instance was successfully created and connected to the EC2 instance. The database was accessed successfully, and basic SQL operations were performed.

LEARNING OUTCOME-

- Learned how to create an AWS RDS MariaDB instance.
- Learned to configure VPC and Security Group for database access.
- Learned to connect EC2 with RDS using the endpoint.
- Learned basic SQL commands for database creation and verification.

THANK YOU
